

Effects of nonstructural carbohydrate level and starch:sugar ratio on microbial metabolism in continuous culture of rumen contents

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Abstract

A dual flow continuous culture system was used to examine the effects of varying sugar as a proportion of sugar plus starch (NSC) at three concentrations of NSC on nutrient digestibility, ruminal fermentation and microbial growth. Diets contained 240, 280 and 330 g NSC/kg DM. At each NSC level, diets contained 29, 63 and 95 g sugar/kg DM. Corn grain and corn silage were the major starch sources. Liquid sugar blend (LSB) was the supplemental sugar source. Negative responses with interactions were seen at the 63 g sugar/kg DM level compared to no added sugar or 95 g sugar/kg DM in microbial growth, dry matter digested, protein digested, and total carbohydrate digested. Interactions resulted mostly because the negative responses to the 63 g sugar/kg DM were with the 240 and 280 g NSC/kg DM levels, but not with the 330 g NSC/kg DM level. Sugar addition depressed fiber digestion only with the 240 g NSC/kg DM level. At 330 g NSC/kg DM, all sugar substitutions were beneficial. Microbial responses were highest on the 330 g NSC/kg DM diet with 95 g sugar/kg DM, in

Abbreviations: ADF, acid detergent fiber; ADL, acid detergent lignin; CP, crude protein; DM, dry matter; LSB, liquid sugar blend; aNDF, NDF, assayed with sodium sulfite and heat-stable amylase; NFC, non-fiber carbohydrate 1000 – (g/kg CP + NDF + ash + fat); NSC, non-structural carbohydrates (g/kg sugar + g/kg starch); NPN, non-protein nitrogen; OM, organic matter; VFA, volatile fatty acids

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which sugar made up 290 g/kg of the total NSC. Sugar additions at 95 g sugar/kg DM resulted in large increases in butyrate production. All sugar levels decreased acetate production but had little effect on propionate production.

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1. Introduction

Sugars and starches provide the major source of carbohydrate for the growth of rumen microbes in diets of high producing dairy cows. Starches usually make up 900 g/kg of the total starch plus sugar. Hall and Herejk (2001) found that sucrose initiated *in vitro* microbial growth most rapidly followed by pectin, starch and isolated neutral detergent fiber (aNDF). With the intent of increasing the utilization of rapidly soluble protein or NPN by rumen microbes, and thus increasing microbial growth, a number of studies have been conducted in which part of the dietary starch has been replaced by sugars. A reduction in ruminal ammonia concentration has been noted in many studies in which sugars were added to the diet (Chamberlain et al., 1985; Giduck and Fontenot, 1987; Sannes et al., 2002; Lee et al., 2003). This suggests a more efficient utilization of the rapidly available nitrogen components in the diet and a concomitant increase in microbial growth and metabolism. However, Vallimont et al. (2004) found that replacing starch with sucrose at 0, 25, 50 and 75 g/kg DM in a continuous culture system resulted in no changes in ruminal ammonia concentrations and no effect on bacterial nitrogen production. McCormick et al. (2001) found that sucrose addition to the diet at 50 g/kg of the DM reduced rumen ammonia concentration but the same amount of lactose added to the diet had no effect. Substitution of starch with sugar also may affect fiber digestion. Sugars added to forage-based diets have been found to decrease fiber digestion (Huhtanen and Khalili, 1991; Heldt et al., 1999). Vallimont et al. (2004) reported, however, that low levels of sugar (25 and 50 g/kg DM) depressed, while 75 g/kg DM increased, aNDF digestion in diets containing 320 g nonstructural carbohydrates (NSC)/kg DM.

In lactation studies involving addition of sugars from molasses, whey, sucrose, dextrose or lactose, responses in dry matter intake and milk production have been inconsistent (Wing et al., 1988; Morales et al., 1989; Maiga et al., 1995). Broderick and Radloff (2004) replaced dietary high moisture corn with incremental amounts of dried molasses (0, 40, 80, and 120 g/kg DM) and found that 35 mg/kg fat-corrected milk responded in a positive quadratic manner (41.2, 42.2, 42.7, and 40.3 kg/d) and was highest when the diet contained 80 g molasses/kg DM.

One possible explanation for the inconsistent production responses to sugars is that because of the high ruminal liquid dilution rates observed shortly after eating (Dehareng and Godeau, 1989; Susmel et al., 1995; Sutoh et al., 1996), a high proportion of the sugar may leave the rumen prior to fermentation and not contribute to microbial end-product formation or microbial protein flow. Data of Henning et al. (1991, 1993) indicate sugar content can persist at high levels in rumen fluid *in vitro* and *in vivo* for several hours after eating. In these studies, the rates of sugar disappearance ranged from 0.19 to 0.69 h⁻¹. Since sugars dissolve rapidly, they could flow from the rumen with the liquid fraction, which may

reach a passage rate of over 0.30 h^{-1} during or shortly after eating (Dehareng and Godeau, 1989).

The objective of this study was to examine the effects of varying sugar as a proportion of sugar and starch (NSC) at three concentrations of NSC on microbial growth and metabolism.

2. Materials and methods

2.1. Diets

The composition and analyses of the diets used are described in Table 1. As with most diets fed to lactating cows in the United States, the diets contained high levels of forages, which ranged from 620 to 820 g/kg dry matter (DM) across all treatments. Because of the high proportion of forage in these diets, a detailed analyses of each forage is provided in Table 2.

Liquid sugar blend (LSB), supplied by PM Ag Products, Inc. and now produced by Westway Feed Products, served as the added sugar source. This is a molasses based product containing 390 g sugars/kg, 30 g crude protein (CP)/kg, 35 g ash/kg, and 670 g dry matter/kg. The major sources of starch in the diets were corn grain and corn silage. Dietary NSC levels were 240, 280 and 330 g/kg DM. At each NSC level, diets contained 29, 63 and 95 g sugar/kg on a DM basis. Feeds were analyzed for DM (Method 930.15; AOAC, 1995), ash (Method 942.05; AOAC, 1995), and ether extract (Method 920.39; AOAC, 1995). Determination of the aNDF and acid detergent fiber (ADF) content in the feed were by the methods of Van Soest et al. (1991) and Method 973.18 AOAC (1995), respectively. The aNDF was determined using sulfite and heat stable amylase and both aNDF and ADF included residual ash. Total nitrogen (N) in feed, effluents, bacterial and ammonia was determined using an automated Tecator digestion system (Tecator, Inc., Herndon, VA, U.S.A.) (Method 976.05; AOAC, 1995). Soluble CP was determined by the method of Krishnamoorthy et al. (1982). The sugars and starches were determined by the procedure of Smith (1969), except that ferricyanide was used to detect reducing sugars. Analysis of volatile fatty acids (VFA) was performed in accordance with the gas chromatographic separation procedure described by Anonymous (1975).

2.2. Continuous culture system

A 12-unit dual flow continuous culture system similar to that described by Hoover et al. (1976) was used. For complex studies of this nature, the use of continuous culture is a practical and appropriate alternative to *in vivo* methods. The dual flow continuous culture technique has been established as a valid method of simulating *in vivo* rumen fermentation (Hannah et al., 1986). The artificial saliva of Weller and Pilgrim (1974) was continuously infused into the fermenter vessels (1163 ml working volume) at a rate to provide a liquid flow of 0.12 h^{-1} . In the dual flow system, a filter is used for removal of liquids and solid particles less than $100 \mu\text{m}$, while undigested feed particles of $100 \mu\text{m}$ and larger flow through an overflow port 19 mm in diameter. The latter represents the flow of solids, and the solids retention time is adjusted by changing the filtrate to overflow ratio. The rate

Table 1
Diet composition and analysis (g/kg dry matter basis)

Ingredient (g/kg)	Diets								
	29 g sugar/kg			63 g sugar/kg			95 g sugar/kg		
	240 g NSC ^a /kg	280 g NSC ^a /kg	330 g NSC ^a /kg	240 g NSC ^a /kg	280 g NSC ^a /kg	330 g NSC ^a /kg	240 g NSC ^a /kg	280 g NSC ^a /kg	330 g NSC ^a /kg
Grass legume silage	486.6	430.0	388.2	466.7	408.6	386.2	455.7	395.7	370.8
Corn silage	333.9	327.0	287.3	321.0	312.1	259.4	295.1	299.0	249.4
Corn, grain (ground, 4 mm)	94.7	151.7	219.3	55.8	115.8	187.4	22.9	76.2	148.3
Dicalcium phosphate	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
Liquid sugar blend	–	–	–	62.8	62.8	62.8	125.6	125.6	125.6
Soybean meal 44, solvent-extracted,	37.9	42.9	57.4	44.9	50.8	54.4	51.8	52.6	54.8
Urea	–	1.6	1.00	–	1.00	1.00	–	1.99	2.19
Protein blend-WP2 ^b	40.9	40.9	40.9	42.9	42.9	42.9	42.9	42.9	42.9
Analysis (g/kg)									
CP ^c	182.7	181.5	179.1	185.2	182.8	175.8	189.6	183.5	184.7
SIP (g/kg CP) ^d	453.4	445.7	434.7	452.6	429.1	423.5	449.4	456.5	448.2
Neutral detergent fiber	398.3	365.8	325.3	372.4	357.0	332.1	374.6	354.4	325.0
Acid detergent fiber	253.0	234.6	215.8	248.3	230.3	215.6	247.4	232.7	217.3
Fat	31.1	30.9	32.2	25.9	27.4	27.3	25.4	26.7	26.7
NSC	239.7	269.1	336.0	239.0	279.9	326.0	240.9	277.2	330.7
Sugar	28.9	29.3	29.1	62.9	62.6	63.3	95.2	95.2	95.2
Starch	210.8	239.8	306.9	176.1	217.3	262.7	145.7	182.0	235.5
Ash	75.2	68.0	68.3	78.5	67.9	61.5	79.9	76.7	76.8
NFC ^e	312.7	353.8	395.1	338.0	364.9	403.3	330.5	358.7	386.8

^a Nonstructural carbohydrates (sugar + starch).

^b Manufactured for Southern States, Coop., Inc. Ingredients: ring dried blood meal, hydrolyzed poultry feathers, fish meal, calcium carbonate, ethoxyquin, processed grain by-products.

^c Crude protein.

^d Soluble intake protein.

^e Non-fiber carbohydrates (1000 – (aNDF, g/kg + CP, g/kg + ash, g/kg + fat, g/kg)).

Table 2

Forage analyses (g/kg dry matter basis), unless otherwise stated

Component	Corn silage	Haylage
Dry matter	326.1	469.0
Crude protein	62.9	199.4
Soluble protein (g/kg crude protein)	499.5	613.0
ADFIP ^a (g/kg crude protein)	34.7	30.4
NDFIP ^b (g/kg crude protein)	153.8	45.7
aNeutral detergent fiber	474.3	415.6
Lignin (sa)	93.2	110.6
Acid detergent fiber	298.8	301.3
NSC ^c	310.4	94.2
Sugar	15.4	34.9
Starch	295.0	59.3
Ash	45.5	99.1
Fat	26.3	30.6
NFC ^d	391.0	255.3
Ca	3.3	8.3
P	2.8	4.5
Mg	1.3	2.7
K	10.4	29.7
S	25.6	24.8
Na	0.4	0.6
Cl	0.01	0.03
Fe (ppm)	34.22	55.71
Zn (ppm)	24.86	28.38
Co (ppm)	0.10	0.11
Mn (ppm)	47.05	45.47
Cu (ppm)	6.47	11.39
I (ppm)	0.51	0.52
Volatile fatty acids		
Acetic	14.8	10.7
Propionic	3.7	0.07
Iso-butyric	0.3	0.1
Butyric	0.3	0.01
Lactic	22.9	60.5
Total VFA	41.9	71.5

^a Acid detergent insoluble protein.^b Neutral detergent insoluble protein.^c Nonstructural carbohydrates (sugar + starch).^d Nonfiber carbohydrates (1000 – (aNDF, g/kg + CP, g/kg + ash, g/kg + fat, g/kg)).

of overflow in this study was adjusted to 0.045 h^{-1} to give a mean solids retention time of 22 h. Each fermenter was inoculated with mixed rumen contents from two ruminally cannulated lactating Holstein cows that were fed a diet similar to the experimental diet with sugar, 29 g/kg, and NSC, 330 g/kg, as shown in Table 1. Fermentation temperatures were maintained at 39 °C and pH was monitored. Daily feed was 100 g DM divided into two portions. All diets were fermented in triplicate. Fermentation periods were 10 days in length. The first 7 days were for equilibration. During the last 3 days, the effluents were collected in an ice bath and a 1 l sample was composited and saved for analysis.

Following collection of the effluent on day 10, the contents of the fermenters were allowed to settle and the upper fluid layer was used for collection of microbes. Two samples (250 ml) were taken from each fermenter and centrifuged for 20 min at $200\times g$. The supernatants were centrifuged for 15 min at $30,000\times g$, the pellets combined, resuspended in saline and again centrifuged for 15 min at $30,000\times g$. The supernatants were discarded and the pellets were resuspended in 20 ml of a 50:50 mixture of distilled water and methanol, and centrifuged for 15 min at $30,000\times g$. The supernatants were poured off and the pellets resuspended in distilled water and lyophilized.

Effluent dry matter was determined by centrifuging a 34–40 g sample of effluent at $30,000\times g$ for 45 min. The supernatant was discarded and the particulate matter was dried at 110°C for 24 h and reweighed. The adaptations of the aNDF and ADF analyses for continuous culture effluents were described by Crawford et al. (1983). Effluent and bacterial concentrations of purines were determined by the procedures of Zinn and Owens (1986). Digestion coefficients for DM, organic matter (OM), aNDF, ADF and CP, and total flows of non-ammonia, microbial and dietary N were calculated as described by Stern and Hoover (1990).

2.3. Statistical analysis

Data were analyzed using the general linear model of SAS (1982). Mean responses were examined for significant differences due to NSC levels, sugar levels and interactions. The responses to sugar and NSC levels also were examined for linear and quadratic effects.

3. Results and discussion

No interactions between levels of NSC and sugar were noted for digestion of DM, ADF, NSC, sugar or starch, so these data are presented as means for NSC (Table 3) and sugar (Table 4). As NSC levels increased, there were trends for linear increases in digestion of NSC ($P=0.10$) and starch ($P=0.14$), with no responses in digestion of DM, ADF or sugar. As level of sugar was increased (Table 4), digestion of DM was affected quadratically, while digestion of ADF and sugar increased linearly ($P=0.08$ and 0.01 , respectively).

Table 3
Effects of level of non-structural carbohydrates (NSC) on digestion

Item	240 g NSC/kg DM	280 g NSC/kg DM	330 g NSC/kg DM	P ^a	
				L	Q
Digestibility					
Dry matter	0.66	0.67	0.66	NS	NS
Acid detergent fiber	0.47	0.49	0.46	NS	NS
Non-structural carbohydrate	0.80	0.84	0.84	0.10	NS
Sugar	0.65	0.63	0.63	NS	NS
Starch	0.84	0.88	0.87	0.14	NS

^a P, probability; L, linear; Q, quadratic.

Table 4
Effects of sugar levels on digestibility

Item	29 g sugar/kg DM	63 g sugar/kg DM	95 g sugar/kg DM	P ^a	
				L	Q
Digestibility					
Dry matter	0.66	0.63	0.70	NS	0.02
Acid detergent fiber	0.46	0.45	0.51	0.08	NS
Non-structural carbohydrate	0.81	0.83	0.84	NS	NS
Sugar	0.45	0.67	0.79	0.01	NS
Starch	0.85	0.87	0.86	NS	NS

^a P, probability; L, linear; Q, quadratic.

Sugar digestion increased with sugar level regardless of NSC content ($P=0.01$) (Table 4). A surprising aspect of sugar and starch digestion was that across all three NSC levels, sugar digestion was less than that of starch, averaging 635 g/kg *versus* 863 g/kg (Table 3). The cause of the low digestion across NSC levels appears related to the availability of the sugar in the natural feed ingredients (primarily the forages and soybean meal) as indicated by the sugar digestion value of only 450 g/kg for the 29 g sugar/kg DM diet *versus* 791 g/kg for the 95 g sugar/kg DM diet (Table 4). The reason for the low level of sugar digested when no LSB was added is not clear; however, sugar digestion was increased considerably by additions of the LSB. Assuming that the level of refractory sugar was the same in all treatments, it was estimated that 890–960 g/kg of the sugar from the LSB was available.

Digestion of aNDF, grams of total carbohydrate digested per day (aNDF + starch + sugar) and protein all were affected by the treatments, but each had a $\text{NSC} \times \text{sugar}$ interaction, indicating that responses to sugar differed at various levels of NSC (Table 5). When no LSB was added, as represented by the 29 g sugar/kg DM level, the aNDF digestion coefficients decreased with increasing NSC. This effect has been shown in a number of studies (Hoover, 1986). Addition of sugar to the low NSC diet reduced fiber digestion; however, fiber digestion was either not affected or increased in the 280 and 330 g NSC/kg DM diets when sugar was added. These data indicated that fiber digestion in high forage diets may not respond to sugar addition as well as diets with 280 g NSC or more per kg DM. When Vallimont et al. (2004) incrementally replaced starch with sucrose (0, 25, 50 and 75 g/kg DM) in a continuous culture system with diets containing 310–330 g NSC/kg DM, aNDF digestibility quadratically increased from 0.61 to 0.66 g/g. Thus in this study as well as that of Vallimont et al. (2004) substituting sugar for starch at levels of 75–95 g sugar/kg DM in diets with more than 240 g NSC/kg DM resulted in increased digestion of NDF. This may explain the increased DM intake as a result of adding sugars to diets fed to cattle (Morales et al., 1989; Wing et al., 1988; Broderick and Radloff, 2004). Total carbohydrate digested, which consists of aNDF + NSC, is a major factor affecting microbial growth. With the exception of the 240 g NSC/kg DM diet, total carbohydrate digested was unaffected by the 63 g sugar/kg DM substitution and enhanced by the 95 g sugar/kg DM substitution. Both NSC and sugar had a quadratic effect on protein digestion with the 280 g NSC/kg DM diets and 63 g sugar/kg DM diets being low. A high level of proteolytic activity was supported with the 330 g NSC/kg DM diet only when some sugar was substituted for starch.

Table 5

Effects of non-structural carbohydrates (NSC) and sugar on digestion of neutral detergent fiber, total carbohydrates and protein

Item	g sugar/ kg DM	240 g NSC/ kg DM	280 g NSC/ kg DM	330 g NSC/ kg DM	Sugar means
Digestion (g/g)					
Neutral detergent fiber ^{a,b}	29	0.50	0.45	0.39	0.45
	63	0.47	0.44	0.45	0.45
	95	0.45	0.52	0.43	0.47
	NSC mean	0.47	0.47	0.42	
Carbohydrate digested ^{a,b} (g/d)	29	40	39	41	40
	63	38	39	41	38
	95	36	42	43	41
	NSC mean	38	40	42	
Protein ^{b,c}	29	0.74	0.73	0.66	0.71
	63	0.67	0.60	0.76	0.68
	95	0.74	0.66	0.80	0.73
	NSC mean	0.72	0.66	0.77	

^a Linear effect; NSC, $P=0.01$.^b Sugar $\times$ NSC interactions, $P\leq 0.05$.^c Quadratic effects: sugar, $P=0.02$; NSC, $P=0.08$.

There were no interactions between NSC and sugar levels in the molar ratios of volatile fatty acids. Although proportions of acetic and iso-valeric acids were slightly increased at the 280 g NSC/kg DM level, no other responses to NSC were found (Table 6). Proportions of acetic acid decreased ($P=0.01$) and butyric acid increased ($P=0.01$) as sugar addition increased (Table 7). These responses in VFA molar ratios are similar to those reported in other studies (Rooke et al., 1987; Wu et al., 1994). In the current study, the major changes in proportions of acetate and butyrate were associated with the highest inclusion of sugar.

Table 6

Effects of non-structural carbohydrate (NSC) on volatile fatty acid (VFA) production, molar ratios and average pH

Item	240 g NSC/kg DM	280 g NSC/kg DM	330 g NSC/kg DM	P ^a	
				L	Q
Total VFA (mmol/d)	380	390	370	NS	NS
VFA (mmol/mol)					
Acetate	565	585	566	NS	0.08
Propionate	240	223	227	NS	NS
Iso-butyrate	10.2	9.8	8.2	NS	NS
Butyrate	143	139	156	NS	NS
Iso-valerate	7.5	10.5	6.7	NS	0.06
Valerate	33.9	33.0	37.9	NS	NS
Mean pH	6.3	6.3	6.3	NS	NS

^a P, probability; L, linear; Q, quadratic.

Table 7

Effects sugar on volatile fatty acid (VFA) production, molar ratios and average pH

Item	29 g sugar/kg DM	63 g sugar/kg DM	95 g sugar/kg DM	P ^a	
				L	Q
Total VFA (mmol/d)	388	372	380	NS	NS
VFA (mmol/mol)					
Acetate	586	583	546	0.01	NS
Propionate	234	238	218	NS	NS
Iso-butyric	11.5	8.3	8.4	NS	NS
Butyric	131	129	179	0.01	NS
Iso-valeric	8.2	7.9	8.5	NS	NS
Valeric	30.0	34.7	40.1	0.01	NS
Mean pH	6.3	6.3	6.3	NS	NS

^a P, probability; L, linear; Q, quadratic.

This finding is in agreement with Vallimont et al. (2004), who reported similar responses to substituting sugar for 75 g/kg DM of starch. In the latter study, a decrease in proportion of propionate in response to added sugar also was reported. Although not statistically significant, a similar trend was seen in this study. In contrast to the report of Vallimont et al. (2004) valerate increased significantly with the increase in sugar. Fermentation pH was not significantly affected by either NSC or sugars and ranged from 5.95 to 6.75 throughout the day.

There were some interactions between NSC and sugar levels in N partitioning (Table 8). Ammonia levels decreased linearly ($P<0.01$) with increasing levels of NSC, regardless of sugar levels, and a positive, linear effect was seen in ammonia in response to increases in sugar ($P=0.02$). Ammonia responses to added sugar have not been consistent. Decreases in ammonia have been reported by several authors (Chamberlain et al., 1985; Giduck and Fontenot, 1987; Sannes et al., 2002; Lee et al., 2003). Vallimont et al. (2004), using a culture system similar to that used in the current study, reported no change in ammonia concentration when up to 75 g/kg DM of the dietary starch was replaced by sucrose. Broderick and Radloff (2004) replaced high moisture corn with incremental amounts of dried molasses (0, 40, 80, and 120 g/kg DM) and found that rumen ammonia concentration in dairy cows responded in a quadratic manner with 16.5, 13.6, 15.7, and 16.1 mg/dl, respectively. The similar tendency for ammonia to increase with increasing sugar in the current study is associated with increased proteolysis (Table 5) coupled with a significant ($P=0.03$) decrease in conversion of digested feed N to microbial N (Table 9), which resulted in the increase seen in ammonia.

The positive quadratic response in rumen undegraded feed nitrogen to NSC level ($P=0.02$) reflects the effects of NSC on protein digestion as shown in Table 5, where a negative quadratic response was shown. Level of sugar also resulted in a positive quadratic effect ($P=0.08$) on rumen undegraded feed nitrogen, again reflecting the reduced rumen proteolysis shown in Table 5. Although there was a sugar $\times$ NSC interaction, in general the highest average protein digestion was associated the highest sugar level.

Table 8

Effects of non-structural carbohydrates (NSC) and sugar on nitrogen partitioning

Item	g sugar/ kg DM	240 g NSC/ kg DM	280 g NSC/ kg DM	330 g NSC/ kg DM	Sugar means
Ammonia N ^a (mg/dl)	29	6.5	7.2	5.0	6.2
	63	7.3	5.8	4.8	6.0
	95	9.0	8.3	5.8	7.7
	NSC mean	7.6	7.1	5.1	
Ruminally undegraded feed N ^{b,c} (g/d)	29	0.83	0.87	1.08	0.93
	63	1.08	1.29	0.75	1.04
	95	0.87	1.09	0.66	0.87
	NSC mean	0.93	1.08	0.83	
Microbial N ^{b,c} (g/d)	29	2.17	2.10	1.91	2.06
	63	1.92	1.74	2.22	1.96
	95	2.15	1.87	2.40	2.14
	NSC mean	2.08	1.90	2.17	
Total NAN ^d (g/d)	29	3.00	2.97	3.00	2.99
	63	3.01	3.03	2.97	3.01
	95	3.03	2.96	3.06	3.02
	NSC mean	3.02	2.99	3.01	

^a Linear effects: NSC, $P < 0.01$; Sugar, $P < 0.02$.^b Quadratic effects: sugar, $P < 0.08$; NSC, $P < 0.02$.^c Sugar $\times$ NSC interaction; $P = 0.02$.^d Non-ammonia nitrogen.

Table 9

Nonstructural carbohydrates (NSC) and sugar on microbial efficiencies

Item	g sugar/ kg DM	240 g NSC/ kg DM	280 g NSC/ kg DM	330 g NSC/ kg DM	Sugar means
Grams microbial ^a (N/kg organic matter digested)	29	46.4	44.0	44.1	44.8
	63	42.6	41.5	44.5	42.8
	95	47.3	40.8	46.3	44.8
	NSC mean	45.4	42.1	45.0	
Grams microbial ^{b,c} (N/kg carbohydrate digested)	29	55.3	54.5	47.8	52.5
	63	52.8	45.0	52.3	50.0
	95	59.6	44.3	56.8	53.6
	NSC mean	55.9	47.9	52.3	
Digested CP N converted to microbial N ^d (g/kg)	29	908	898	920	909
	63	886	899	934	906
	95	876	870	926	890
	NSC mean	890	889	926	

^a Quadratic NSC effect, $P = 0.04$.^b Quadratic NSC effect, $P = 0.01$.^c Sugar $\times$ NSC interaction, $P = 0.03$.^d Linear effects: sugar, $P = 0.03$; NSC, $P = 0.01$.

The microbial growth, expressed as g microbial N produced per day (Table 8), responded quadratically to both increased NSC and, to some extent, increased sugar. There was, however a significant ($P=0.02$) interaction between the two treatments, in that replacing starch with sugar appeared more beneficial when done with the 330 g NSC/kg DM diet than with the 240 or 280 g NSC/kg DM diets. The sources of carbohydrates available may have been responsible for the microbial growth responses. Hall and Herejk (2001) reported that starches resulted in more microbial growth than did sugar. In the current study, the digestibility of sugar increased with sugar level (Table 4), thus the proportion of fermented NSC from sugar increased both with added sugar level and with decreased NSC level. In the 280 g NSC/kg DM diet sugar made up a higher proportion of digestible NSC than on the 330 g NSC/kg DM diet and microbial responses were lower. On the high NSC diet, sugar made up a lower proportion of the NSC and microbial responses were not pronounced. Vallimont et al. (2004) found no increase in microbial growth when up to 75 g/kg starch was replaced by sugar, but the digestibility of the added sugar was not presented. It may be that microbial growth is stimulated by the ratio of digestible sugar, digestible starch and digestible aNDF (which decreased with increasing NSC level).

Efficiency of microbial growth is expressed as g microbial N produced per kg of carbohydrate or organic matter digested (Table 9). For the 280 g NSC/kg DM diets, microbial N production per kg of carbohydrate digested was significantly lower ($P=0.01$), especially when sugars were added to the diets ($P=0.03$). Microbial N produced per kg OM digested also responded negatively in a quadratic manner to NSC level ($P=0.04$), with sugar having no effect.

Protein metabolism is a component that should be considered in studies involving starch and sugar. Microbial efficiency will decline if fermentation is uncoupled due to lack of rumen degradable protein (Russell and Cook, 1995; Van Kessel and Russell, 1996; Bond et al., 1998). Based on the protein digestion coefficients in Table 5, it is apparent that most of the low microbial efficiency values in Table 9 are associated with low protein digestion coefficients. Efficiency of use of digested CP for microbial N was linearly decreased by sugar addition to the diet ($P=0.03$) but linearly increased with NSC ($P=0.01$).

4. Conclusions

Digestion of sugar increased linearly with increasing additions of sugar from liquid sugar blend, but did not exceed that of starch digestion at the highest sugar level used in this study. As a consequence, replacement of starch with sugar was ineffective in providing more available carbohydrate than the replaced starch. Substitution of a small amount of sugar from LSB (63 g sugar/kg DM) for starch in diets with only 240–280 g NSC/kg DM resulted in several negative microbial responses. For this reason, when less than 95 g sugar/kg DM is included in diets with up to 280 g NSC/kg DM, it probably should be added to the diets and not substituted for part of the starch. Data support the theory that inconsistent production responses to sugar may be due to ruminal escape of sugar. The low digestibility of the sugars contained in the feeds was unexpected and bears further investigation.

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